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Binding machine

Abstract

A binding machine includes a wire feeding unit that feeds a wire, a curl forming unit that forms a path along which the wire fed by the wire feeding unit is to be wound around an object, a cutting unit that cuts the wire wound on the object, a binding unit that twists the wire wound on the object and cut by the cutting unit, and a drive unit that drives the binding unit. The drive unit includes a motor and a decelerator that perform deceleration and amplification of torque. The decelerator includes a sun gear, a planetary gear in mesh with the sun gear, a planet cage that supports the planetary gear, and an internal gear in mesh with the planetary gear. The planet cage is rotatably supported on one side and the other side along an axis direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application claims priority from Japanese Patent Application No. 2021-171966 filed on Oct. 20, 2021, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD

(2) The present invention relates to a binding machine configured to bind a to-be-bound object such as a reinforcing bar with a wire.

BACKGROUND ART

(3) For concrete buildings, reinforcing bars are used so as to improve strength. The reinforcing bars are bound with wires so that the reinforcing bars do not deviate from predetermined positions during concrete placement.

(4) In the related art, suggested is a binding machine referred to as a reinforcing bar binding machine configured to wind a wire on two or more reinforcing bars and to twist the wire wound on the reinforcing bars, thereby binding the two or more reinforcing bars with the wire.

(5) The reinforcing bar binding machine includes a wire feeding unit configured to feed a wire, a curl forming unit configured to form a path along which the wire is to be wound around a to-be-bound object, a cutting unit configured to cut the wire, a binding unit configured to twist the wire, and a drive unit configured to drive the binding unit, in which the drive unit includes a motor and a decelerator configured to perform deceleration and amplification of torque. The decelerator is configured by a planet gear in which an input shaft and an output shaft are coaxially arrayed, and includes a sun gear attached to the input shaft, a planetary gear in mesh with the sun gear, a planet cage configured to support the planetary gear, and an internal gear in mesh with the planetary gear (for example, refer to JP2018-108850A).

(6) As for the decelerator used in the binding machine, one side of the planet cage along an axis direction is rotatably supported by a main body part to which the internal gear is fixed. In contrast, the other side of the planet cage along the axis direction is in mesh with the planetary gear and the internal gear.

(7) Since a predetermined clearance is required for the mesh between gears, there is a clearance in the mesh between the planetary gear and the internal gear, and there is a possibility that the planet cage will be inclined with respect to the axis direction.

(8) The present invention has been made so as to solve the problem, and an object thereof is to provide a binding machine capable of suppressing a planet cage configured to support a planetary gear in a planet gear from being inclined.

SUMMARY

(9) According to an aspect of the present invention, there is provided a binding machine including a wire feeding unit configured to feed a wire, a curl forming unit configured to form a path along which the wire fed by the wire feeding unit is to be wound around an object, a cutting unit configured to cut the wire wound on the object, a binding unit configured to twist the wire wound on the object and cut by the cutting unit, and a drive unit configured to drive the binding unit. The drive unit comprises a motor and a decelerator configured to perform deceleration and amplification of torque. The decelerator includes a sun gear, a planetary gear in mesh with the sun gear, a planet cage configured to support the planetary gear, and an internal gear in mesh with the planetary gear. The planet cage is rotatably supported on one side and the other side along an axis direction.

(10) According to an another aspect of the present invention, there is provided a binding machine including a wire feeding unit configured to feed a wire, a curl forming unit configured to form a path along which the wire fed by the wire feeding unit is to be wound around an object, a cutting unit configured to cut the wire wound on the object, a binding unit configured to twist the wire wound on the object and cut by the cutting unit, and a drive unit configured to drive the binding unit. The drive unit includes a motor and a decelerator configured to perform deceleration and

amplification of torque. The decelerator includes a first sun gear attached to a shaft of the motor, a first planetary gear in mesh with the first sun gear, a first planet cage configured to support the first planetary gear, a second sun gear provided to the first planet cage, a second planetary gear in mesh with the second sun gear, and a second planet cage configured to support the second planetary gear. The second planet cage is rotatably supported on one side and the other side along an axis direction.
(11) In the present invention, the decelerator is configured by the planet gear in which the input shaft and the output shaft are coaxially arrayed, and the planet cage configuring the output shaft is rotatably supported on one side and the other side along the axis direction.
(12) According to the present invention, with a simple configuration, the planet cage is suppressed from being inclined with respect to the axis direction, and changes in meshes between the sun gear and the planetary gear and between the planetary gear and the internal gear, and interferences between gears aligned in the axis direction, between a gear and a planet cage, and the like can be suppressed.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is an internal configuration view showing an example of an overall configuration of a reinforcing bar binding machine of the present embodiment, as seen from a side.
(2) FIG. **2**A is an internal configuration view showing an example of a main part configuration of the reinforcing bar binding machine of the present embodiment, as seen from a side.
(3) FIG. **2**B is an internal configuration view showing the example of the main part configuration of the reinforcing bar binding machine of the present embodiment, as seen from a side.
(4) FIG. **2**C is an internal configuration view showing the example of the main part configuration of the reinforcing bar binding machine of the present embodiment, as seen from a side.
(5) FIG. **3**A is a plan view showing an example of a binding unit according to the present embodiment.
(6) FIG. **3**B is a plan view showing the example of the binding unit according to the present embodiment.
(7) FIG. **3**C is a plan view showing the example of the binding unit according to the present embodiment.
(8) FIG. **3**D is a plan view of main parts showing a modified embodiment of the binding unit according to the present embodiment.
(9) FIG. **3**E is a plan view of main parts showing a modified embodiment of the binding unit according to the present embodiment.
(10) FIG. **3**F is a plan view of main parts showing a modified embodiment of the binding unit according to the present embodiment.
(11) FIG. **4**A is a plan view showing an example of a cutting unit according to the present embodiment.
(12) FIG. **4**B is a plan view showing the example of the cutting unit according to the present embodiment.
(13) FIG. **4**C is a perspective view showing the example of the cutting unit of the present embodiment.
(14) FIG. **4**D is a perspective view showing the example of the cutting unit of the present embodiment.
(15) FIG. **4**E is a perspective view showing the example of the cutting unit of the present embodiment.
(16) FIG. **4**F is a plan view showing a modified embodiment of the cutting unit according to the

present embodiment.
(17) FIG. **4**G is a plan view showing a modified embodiment of the cutting unit according to the present embodiment.
(18) FIG. **5**A is a side cross-sectional view showing an example of a decelerator according to the present embodiment.
(19) FIG. **5**B is a perspective view showing the example of the decelerator according to the present embodiment.
(20) FIG. **5**C is a side cross-sectional view of main parts showing a modified embodiment of the decelerator according to the present embodiment.
(21) FIG. **5**D is a perspective view showing the modified embodiment of the decelerator according to the present embodiment.
(22) FIG. **6**A is a plan view showing an example of a curl forming unit according to the present embodiment.
(23) FIG. **6**B is a plan view showing the example of the curl forming unit according to the present embodiment.
(24) FIG. **6**C is a plan view showing the example of the curl forming unit according to the present embodiment.
(25) FIG. **6**D is a plan view showing the example of the curl forming unit according to the present embodiment.
(26) FIG. **7**A is an operation explanatory diagram showing an example of operations of the binding unit, a transmission unit and the cutting unit according to the present embodiment.
(27) FIG. **7**B is an operation explanatory diagram showing the example of operations of the binding unit, the transmission unit and the cutting unit according to the present embodiment.
(28) FIG. **7**C is an operation explanatory diagram showing the example of operations of the binding unit, the transmission unit and the cutting unit according to the present embodiment.
(29) FIG. **7**D is an operation explanatory diagram showing the example of operations of the binding unit, the transmission unit and the cutting unit according to the present embodiment.
(30) FIG. **7**E is an operation explanatory diagram showing the example of operations of the binding unit, the transmission unit and the cutting unit according to the present embodiment.
(31) FIG. **7**F is an operation explanatory diagram showing the example of operations of the binding unit, the transmission unit and the cutting unit according to the present embodiment.
(32) FIG. **7**G is an operation explanatory diagram showing the example of operations of the binding unit, the transmission unit and the cutting unit according to the present embodiment.
(33) FIG. **8**A is a side view showing a modified embodiment of the transmission unit according to the present embodiment.
(34) FIG. **8**B is a side view showing the modified embodiment of the transmission unit according to the present embodiment.
(35) FIG. **8**C is a side view showing the modified embodiment of the transmission unit according to the present embodiment.
(36) FIG. **9**A is a side cross-sectional view showing the modified embodiment of the transmission unit according to the present embodiment.
(37) FIG. **9**B is a side cross-sectional view showing the modified embodiment of the transmission unit according to the present embodiment.
(38) FIG. **9**C is a side cross-sectional view of parts showing the modified embodiment of the transmission unit according to the present embodiment.
DESCRIPTION OF EMBODIMENTS
(39) Hereinafter, an example of a reinforcing bar binding machine as an embodiment of the binding machine of the present invention will be described with reference to the drawings.
(40) <Overall Configuration Example of Reinforcing Bar Binding Machine of Present Embodiment>

(41) FIG. **1** is an internal configuration view showing an example of an overall configuration of a reinforcing bar binding machine of the present embodiment, as seen from a side.
(42) A reinforcing bar binding machine **1**A is configured to feed a wire W in a forward direction denoted with an arrow F, to wind the wire around reinforcing bars S, which are a to-be-bound object (an object), to feed the wire W wound around the reinforcing bars S in a reverse direction denoted with an arrow R, to wind the wire on the reinforcing bars S, to cut the wire, and to twist the wire W, thereby binding the reinforcing bars S with the wire W.
(43) The reinforcing bar binding machine **1**A includes a magazine **2** in which the wire W is accommodated, a wire feeding unit **3** configured to feed the wire W, and a wire guide **4** configured to guide the wire W, so as to implement the above-described functions. In addition, the reinforcing bar binding machine **1**A includes a curl forming unit **5** configured to form a path along which the wire W fed by the wire feeding unit **3** is to be wound around the reinforcing bars S, and a cutting unit **6** configured to cut the wire W wound on the reinforcing bars S. Further, the reinforcing bar binding machine **1**A includes a binding unit **7** configured to twist the wire W wound on the reinforcing bars S, a drive unit **8** configured to drive the binding unit **7**, and a transmission unit **9** configured to transmit an operation of the binding unit **7** to the cutting unit **6**.
(44) Further, the reinforcing bar binding machine **1**A has such a form that an operator grips and uses with a hand, and has a main body part **10** and a handle part **11**.
(45) The magazine **2** is an example of the accommodation unit, and a reel **20** on which the long wire W is wound to be reeled out is rotatably and detachably accommodated therein. For the wire W, a wire made of a plastically deformable metal wire, a wire having a metal wire covered with a resin, or a twisted wire is used.
(46) In a configuration in which the reinforcing bars S are bound with one wire W, one wire W is wound on a hub part (not shown) of the reel **20**, and one wire W can be pulled out while the reel **20** rotates. In addition, in a configuration in which the reinforcing bars S are bound with a plurality of wires W, the plurality of wires W are wound on the hub part, and the plurality of wires W can be pulled out at the same time while the reel **20** rotates. For example, in a configuration in which the reinforcing bars S are bound with two wires W, the two wires W are wound on the hub part, and the two wires W can be pulled out at the same time while the reel **20** rotates.
(47) The wire feeding unit **3** includes a pair of feeding gears **30** configured to sandwich and feed the wire W. The wire feeding unit **3** is configured such that a rotating operation of a feeding motor (not shown) is transmitted to rotate the feeding gears **30**. Thereby, the wire feeding unit **3** is configured to feed the wire W sandwiched between the pair of feeding gears **30** along an extension direction of the wire W. In a configuration in which a plurality of, for example, two wires W are fed to bind the reinforcing bars S, the two wires W are fed aligned in parallel.
(48) The wire feeding unit **3** is configured such that a rotation direction of the feeding motor (not shown) is switched between forward and reverse directions to switch rotation directions of the feeding gears **30**, thereby feeding the wire W in the forward direction denoted with the arrow F, feeding the wire W in the reverse direction denoted with the arrow R, or switching the feeding direction of the wire W between the forward and reverse directions.
(49) The wire guide **4** is provided at a predetermined position on an upstream side and a downstream side of the wire feeding unit **3** with respect to a feeding direction of feeding the wire W in the forward direction, respectively. In the configuration in which the two wires W are fed to bind the reinforcing bars S, the wire guide **4** provided on the upstream side of the wire feeding unit **3** is configured to regulate the two wires W in a radial direction, to align the two introduced wires W in parallel and to guide the wires between the pair of feeding gears **30**. The wire guide **4** provided on the downstream side of the wire feeding unit **3** is configured to regulate the two wires W in the radial direction, to align the two introduced wires W in parallel, and to guide the wires toward the cutting unit **6** and the curl forming unit **5**.
(50) The curl forming unit **5** includes a curl guide **50** configured to curl the wire W that is fed by

the wire feeding unit **3**, and an induction guide **51** configured to guide the wire W curled by the curl guide **50** toward the binding unit **7**. In the reinforcing bar binding machine **1**A, the path of the wire W that is fed by the wire feeding unit **3** is regulated by the curl forming unit **5**, so that a locus of the wire W becomes a loop Ru as shown with a dashed-two dotted line in FIG. **1** and the wire W is thus wound around the reinforcing bars S.
(51) In the reinforcing bar binding machine **1**A, the curl guide **50** and the induction guide **51** of curl forming unit **5** are provided at an end portion on a front side of the main body part **10**.
(52) The cutting unit **6** includes a fixed blade part **60** and a movable blade part **61** configured to cut the wire W in cooperation with the fixed blade part **60**. The cutting unit **6**A is configured to cut the wire W by a rotating operation of the movable blade part **61** about the fixed blade part **60** as a fulcrum shaft. In the present specification, the cutting unit **6** is described as the fixed blade part **60** and the movable blade part **61** configured to rotate about the fixed blade part **60** as a fulcrum shaft. However, the movable blade part **61** may be of a slide type configured to linearly slide, not to rotate.
(53) The transmission unit **9** includes a cam **90** configured to rotate by an operation of the binding unit **7**, and a link **91** configured to connect the cam **90** and the movable blade part **61**. The transmission unit **9** is configured to transmit the operation of the binding unit **7** to the movable blade part **61** of the cutting unit **6** via the cam **90** and the link **91**.
(54) The binding unit **7** includes a locking member **70** configured to lock the wire W, and a sleeve **71** configured to actuate the locking member **70**. The drive unit **8** includes a motor **80**, and a decelerator **81** configured to perform deceleration and amplification of torque.
(55) The binding unit **7** is configured to be driven by the drive unit **8**, whereby the sleeve **71** actuates the locking member **70** to lock the wire W. In addition, the binding unit **7** is configured to bind the reinforcing bars S by twisting the wire W after cutting the wire W by the cutting unit **6** in conjunction with the operation of the sleeve **71**.
(56) In the reinforcing bar binding machine **1**A, the wire feeding unit **3**, the wire guide **4**, the cutting unit **6**, the binding unit **7**, the drive unit **8**, the transmission unit **9**, and the like are accommodated within the main body part **10**. In the reinforcing bar binding machine **1**A, the binding unit **7** is provided inside a front side of the main body part **10**, and the drive unit **8** is provided inside a rear side. In addition, in the reinforcing bar binding machine **1**A, a butting portion **16** against which the reinforcing bars S are to be butted is provided at an end portion on the front side of the main body part **10** and between the curl guide **50** and the induction guide **51**.
(57) Further, in the reinforcing bar binding machine **1**A, the handle part **11** extends downward from the main body part **10**, and a battery **15** is detachably mounted to a lower part of the handle part **11**. In addition, in the reinforcing bar binding machine **1**A, the magazine **2** is provided in front of the handle part **11**.
(58) In the reinforcing bar binding machine **1**A, a trigger **12** is provided on a front side of the handle part **11**, and a switch **13** is provided inside the handle part **11**. In the reinforcing bar binding machine **1**A, a control unit **14** is configured to control the motor **80** and a feeding motor (not shown), in response to a state of the switch **13** that is pressed by an operation on the trigger **12**.
(59) <Configuration Example of Main Parts of Reinforcing Bar Binding Machine of Present Embodiment>
(60) FIGS. **2**A to **2**C are internal configuration views showing an example of a main part configuration of the reinforcing bar binding machine of the present embodiment, as seen from a side, in which FIG. **2**A mainly shows the binding unit **7**, the cutting unit **6** and the transmission unit **9**, FIG. **2**B is a cross-sectional view of the cutting unit **6** and the transmission unit **9** in FIG. **2**A, and FIG. **2**C shows the internal configuration by showing an outer shape of the sleeve **71** in FIG. **2**A with a dashed-two dotted line. In addition, FIGS. **3**A to **3**C are plan views showing an example of the binding unit of the present embodiment, and FIGS. **3**D to **3**F are plan views of main parts showing modified embodiments of the binding unit of the present embodiment.

(61) Example of Embodiment of Binding Unit

(62) Next, an example of the binding unit of the present embodiment will be described with reference to each drawing. The binding unit **7** has a rotary shaft **72** configured to move and rotate the sleeve **71**, thereby actuating the locking member **70**. The binding unit **7** and the drive unit **8** are configured such that the rotary shaft **72** and the motor **80** are connected via the decelerator **81** and the rotary shaft **72** is driven by the motor **80** via the decelerator **81**.

(63) The locking member **70** includes a center hook **70**C connected to the rotary shaft **72**, and a first side hook **70**R and a second side hook **70**L configured to open/close with respect to the center hook **70**C.

(64) In the binding unit **7**, a side on which the center hook **70**C, the first side hook **70**R and the second side hook **70**L are provided is referred to as a front side, and a side on which the rotary shaft **72** is connected to the decelerator **81** is referred to as a rear side.

(65) The center hook **70**C is connected to a front end of the rotary shaft **72**, which is one end portion, via a configuration that can rotate with respect to the rotary shaft **72**, can rotate integrally with the rotary shaft **72** and can move integrally with the rotary shaft **72** in an axis direction.

(66) A tip end side of the first side hook **70**R, which is one end portion along the axis direction of the rotary shaft **72**, is located on one side part with respect to the center hook **70**C. In addition, a rear end side of the first side hook **70**R, which is the other end portion along the axis direction of the rotary shaft **72**, is rotatably supported to the center hook **70**C by a shaft **71***b*.

(67) A tip end side of the second side hook **70**L, which is one end portion along the axis direction of the rotary shaft **72**, is located on the other side part with respect to the center hook **70**C. In addition, a rear end side of the second side hook **70**L, which is the other end portion along the axis direction of the rotary shaft **72**, is rotatably supported to the center hook **70**C by the shaft **71***b*.

(68) Thereby, the locking member **70** is configured to open/close in directions in which the tip end side of the first side hook **70**R is contacted/separated with respect to the center hook **70**C by a rotating operation about the shaft **71***b* as a fulcrum. The locking member is also configured to open/close in directions in which the tip end side of the second side hook **70**L is contacted/separated with respect to the center hook **70**C.

(69) The rotary shaft **72** is connected at a rear end, which is the other end portion, to the decelerator **81** via a connection portion **72***b* having a configuration of enabling the rotary shaft **72** to rotate integrally with the decelerator **81** and to move in the axis direction with respect to the decelerator **81**. The connection portion **72***b* has a spring **72***c* for urging backward the rotary shaft **72** toward the decelerator **81**, and regulating a position of the rotary shaft **72** along the axis direction. Thereby, the rotary shaft **72** is configured to be movable forward away from the decelerator **81** while receiving a force pushed backward by the spring **72***c*. Accordingly, the rotary shaft **72** and the locking member **70** connected to the rotary shaft **72** can move forward up to a predetermined amount defined by the connection portion **72***b* while receiving the force pushed backward by the spring **72***c*.

(70) The sleeve **71** has such a shape that a range of a predetermined length along the axis direction of the rotary shaft **72** from an end portion in the forward direction denoted with the arrow A**1** is divided into two in a radial direction and the first side hook **70**R and the second side hook **70**L enter. In addition, the sleeve **71** is formed in a cylindrical shape configured to cover around the rotary shaft **72**, and has a convex portion (not shown) protruding from an inner peripheral surface of a cylinder-shaped space in which the rotary shaft **72** is inserted, and the convex portion enters a groove portion of a feeding screw **72***a* formed along the axis direction on an outer periphery of the rotary shaft **72**.

(71) When the rotary shaft **72** rotates, the sleeve **71** is moved in a front and rear direction along the axis direction of the rotary shaft **72** according to a rotation direction of the rotary shaft **72** by an action of the convex portion (not shown) and the feeding screw **72***a* of the rotary shaft **72**. In addition, when the sleeve **71** is moved to a forward end portion of the feeding screw **72***a* along the axis direction of the rotary shaft **72**, the sleeve is rotated integrally with the rotary shaft **72**.

(72) The sleeve **71** has an opening/closing pin **71***a* configured to open/close the first side hook **70**R and the second side hook **70**L. The first side hook **70**R has an opening/closing guide hole **73**R into which the opening/closing pin **71***a* is inserted, and the second side hook **70**L has an opening/closing guide hole **73**L into which the opening/closing pin **71***a* is inserted.
(73) The opening/closing guide holes **73**R and **73**L are configured by grooves extending along a moving direction of the sleeve **71**. The opening/closing guide hole **73**R has an opening/closing portion **73***a* having a shape of converting linear motion of the opening/closing pin **71***a* configured to move in conjunction with the sleeve **71** into an opening/closing operation by rotation of the first side hook **70**R about the shaft **71***b* as a fulcrum. In addition, the opening/closing guide hole **73**L has an opening/closing portion **73***a* having a shape of converting linear motion of the opening/closing pin **71***a* configured to move in conjunction with the sleeve **71** into an opening/closing operation by rotation of the second side hook **70**L about the shaft **71***b* as a fulcrum. The opening/closing portion **73***a* is configured by a groove inclined with respect to the moving direction of the sleeve **71** and the opening/closing pin **71***a*.
(74) When the sleeve **71** is moved forward (denoted with the arrow A**1**) in a state where the first side hook **70**R is opened with respect to the center hook **70**C, the first side hook **70**R is pushed by the opening/closing pin **71***a*, on an inner wall surface of the opening/closing portion **73***a* formed in the opening/closing guide hole **73**R with respect to a direction in which the first side hook **70**R is closed. Thereby, the first side hook **70**R is rotated about the shaft **71***b* as a fulcrum and is moved toward the center hook **70**C as denoted with the arrow H**1**.
(75) When the sleeve **71** is moved backward (denoted with the arrow A**2**) in a state where the first side hook **70**R is closed with respect to the center hook **70**C, the first side hook **70**R is pushed by the opening/closing pin **71***a*, on an outer wall surface of the opening/closing portion **73***a* formed in the opening/closing guide hole **73**R with respect to a direction in which the first side hook **70**R is opened. Thereby, the first side hook **70**R is rotated about the shaft **71***b* as a fulcrum and is moved away from the center hook **70**C as denoted with the arrow H**2**.
(76) When the sleeve **71** is moved forward (denoted with the arrow A**1**) in a state where the second side hook **70**L is opened with respect to the center hook **70**C, the second side hook **70**L is pushed by the opening/closing pin **71***a*, on an inner wall surface of the opening/closing portion **73***a* formed in the opening/closing guide hole **73**L with respect to a direction in which the second side hook **70**L is closed. Thereby, the second side hook **70**L is rotated about the shaft **71***b* as a fulcrum and is moved toward the center hook **70**C as denoted with the arrow H**1**.
(77) When the sleeve **71** is moved backward (denoted with the arrow A**2**) in a state where the second side hook **70**L is closed with respect to the center hook **70**C, the second side hook **70**L is pushed by the opening/closing pin **71***a*, on an outer wall surface of the opening/closing portion **73***a* formed in the opening/closing guide hole **73**L with respect to a direction in which the second side hook **70**L is opened. Thereby, the second side hook **70**L is rotated about the shaft **71***b* as a fulcrum and is moved away from the center hook **70**C as denoted with the arrow H**2**.
(78) The opening/closing guide hole **73**L provided in the second side hook **70**L has a locking portion **73***b* and an unlocking portion **73***c*. The opening/closing guide hole **73**L is formed with the locking portion **73***b* on a downstream side of the opening/closing portion **73***a* and is formed with the unlocking portion **73***c* on a downstream side of the locking portion **73***b*, with respect to the forward moving direction of the sleeve **71** denoted with the arrow A**1**.
(79) The locking portion **73***b* is formed on the inner wall surface of the opening/closing guide hole **73**L facing toward the direction of the arrow H**1**, which is the direction in which the second side hook **70**L is closed. The locking portion **73***b* faces the outer wall surface of the opening/closing guide hole **73**L with a dimension substantially equivalent to a diameter of the opening/closing pin **71***a*, and extends in parallel to the outer wall surface.
(80) The unlocking portion **73***c* is configured by providing the inner wall surface of the opening/closing guide hole **73**L with a concave portion that is concave with respect to the lock

portion **73***b*. The unlocking portion **73***c* faces the outer wall surface of the opening/closing guide hole **73**L with a dimension slightly greater than the diameter of the opening/closing pin **71***a*, and extends in parallel to the outer wall surface.

(81) As shown in FIG. **3**B, the second side hook **70**L is configured to lock the wire W in a state in which it does not allow movement of the wire W within a range in which the opening/closing pin **71***a* is located at the locking portion **73***b* of the opening/closing guide hole **73**L. Here, within the range in which the opening/closing pin **71***a* is located at the locking portion **73***b* of the opening/closing guide hole **73**L, operations of feeding the wire W in the reverse direction and winding the wire on the reinforcing bars S are performed, as described later.

(82) On the other hand, within a range in which the opening/closing pin **71***a* is moved in the direction of the arrow A**1** in conjunction with the sleeve **71** and the opening/closing pin **71***a* is located at the unlocking portion **73***c* of the opening/closing guide hole **73**L, as shown in FIG. **3**C, the second side hook **70**L becomes movable in a direction of the arrow H**2** in which the second side hook **70**L is spaced apart from the center hook **70**C by such a predetermined amount that the wire W does not come off between the second side hook **70**L and the center hook **70**C.

(83) The sleeve **71** has a bending portion **71***c***1** configured to form the wire W into a predetermined shape by pushing and bending a tip end side of the wire W, which is one end portion, in a predetermined direction. In addition, the sleeve **71** has a bending portion **71***c***2** configured to form the wire W into a predetermined shape by pushing and bending a terminal end side, which is the other end portion of the wire W cut by the cutting unit **6**, in a predetermined direction. The bending portion **71***c***1** and the bending portion **71***c***2** are formed at an end portion of the sleeve **71** in the forward direction denoted with the arrow A**1**.

(84) The sleeve **71** is moved in the forward direction denoted with the arrow A**1**, so that the tip end side of the wire W locked by the center hook **70**C and the second side hook **70**L is pushed and bent toward the reinforcing bars S by the bending portion **71***c***1**. In addition, the sleeve **71** is moved in the forward direction denoted with the arrow A**1**, so that the terminal end side of the wire W locked by the center hook **70**C and the first side hook **70**R and cut by the cutting unit **6** is pushed and bent toward the reinforcing bars S by the bending portion **71***c***2**.

(85) The binding unit **7** includes a rotation regulation part **74** configured to regulate rotations of the locking member **70** and the sleeve **71** in conjunction with the rotating operation of the rotary shaft **72**. The rotation regulation part **74** has a rotation regulation blade **74***a* provided to the sleeve **71**, and a rotation regulation claw (not shown) to which the rotation regulation blade **74***a* is locked and which is provided to the main body part **10**.

(86) The rotation regulation blade **74***a* is configured by a plurality of convex portions protruding radially from an outer periphery of the sleeve **71** and provided with predetermined intervals in a circumferential direction of the sleeve **71**. The rotation regulation blade **74***a* is fixed to the sleeve **71** and is configured to move and rotate integrally with the sleeve **71**.

(87) In an operation area in which the wire W is locked by the locking member **70**, the wire W is wound on the reinforcing bars S and is cut and further the wire W is bent and shaped by the bending portions **71***c***1** and **71***c***2** of the sleeve **71**, the rotation regulation blade **74***a* of the rotation regulation part **74** is locked. When the rotation regulation blade **74***a* is locked, the rotation of the sleeve **71** in conjunction with the rotation of the rotary shaft **72** is regulated, so that the sleeve **71** is moved in the front and rear direction by the rotating operation of the rotary shaft **72**.

(88) In addition, in an operation area in which the wire W locked by the locking member **70** is twisted, the rotation regulation blade **74***a* of the rotation regulation part **74** is unlocked. When the rotation regulation blade **74***a* is unlocked, the sleeve **71** is rotated in conjunction with the rotation of the rotary shaft **72**. The center hook **70**C, the first side hook **70**R and the second side hook **70**L of the locking member **70** locking the wire W are rotated in conjunction with the rotation of the sleeve **71**. In an operation region of the sleeve **71** and the locking member **70** along the axis direction of the rotary shaft **72**, an operation region in which the wire W is locked by the locking

member **70** is referred to as a first operation area. In addition, an operation area in which the wire W locked by the locking member **70** is twisted is referred to as a second operation area.

(89) The binding unit **7** includes a moving member **75** configured to actuate the transmission unit **9**. The moving member **75** is rotatably attached to the sleeve **71**, and is configured not to operate in conjunction with the rotation of the sleeve **71** and to be movable in the front and rear direction in conjunction with the sleeve **71**.

(90) The moving member **75** has an engaging portion **75***a* configured to engage with the cam **90** of the transmission unit **9**. The engaging portion **75***a* is configured not to operate in conjunction with the rotation of the sleeve **71**, and to move in the front and rear direction in conjunction with the sleeve **71**.

(91) Note that, as a modified embodiment of the opening/closing guide hole **73**L provided in the second side hook **70**L, in a modified embodiment shown in FIG. **3**D, the opening/closing guide hole **73**L may be configured to have a first locking portion **73***b*, an unlocking portion **73***c*, and a second locking portion **73***d*. The opening/closing guide hole **73**L is formed with the first locking portion **73***b* on a downstream side of the opening/closing portion **73***a*, the unlocking portion **73***c* on a downstream side of the first locking portion **73***b*, and the second locking portion **73***d* on a downstream side of the unlocking portion **73***c*, with respect to the forward moving direction of the sleeve **71** denoted with the arrow A**1**.

(92) The first locking portion **73***b* and the second locking portion **73***d* are formed in the inner wall surface of the opening/closing guide hole **73**L facing toward the direction of the arrow H**1**, which is the direction in which the second side hook **70**L is closed. The first locking portion **73***b* and the second locking portion **73***d* are configured to face the outer wall surface of the opening/closing guide hole **73**L with a dimension substantially equivalent to the diameter of the opening/closing pin **71***a*, and extend in parallel to the outer wall surface.

(93) The unlocking portion **73***c* is configured by providing the inner wall surface of the opening/closing guide hole **73**L with a concave portion that is concave with respect to the first locking portion **73***b* and the second locking portion **73***b*. The unlocking portion **73***c* is configured to face the outer wall surface of the opening/closing guide hole **73**L with a dimension slightly greater than the diameter of the opening/closing pin **71***a*, and extends in parallel to the outer wall surface.

(94) In the modified embodiment shown in FIG. **3**D, the second side hook **70**L is configured to enable the opening/closing pin **71***a* to move along the inner wall surface of the opening/closing guide hole **73**L by an operation of the opening/closing pin **71***a* moving in the direction of the arrow A**1**, and to lock the wire W in a state in which the wire W is not allowed to move, within a range in which the opening/closing pin **71***a* is located at the first locking portion **73***b* of the opening/closing guide hole **73**L, as shown with the solid line.

(95) On the other hand, within a range in which the opening/closing pin **71***a* is moved in the direction of the arrow A**1** and the opening/closing pin **71***a* is located at the unlocking portion **73***c* of the opening/closing guide hole **73**L, as shown with the dashed-two dotted line, the opening/closing guide hole **73**L can be displaced up to a position denoted with the dashed-two dotted line, with respect to the opening/closing pin **71***a*, and the second side hook **70**L becomes movable in the direction of the arrow H**2** in which the second side hook **70**L is spaced apart from the center hook **70**C by such a predetermined amount that the wire W does not come off between the second side hook **70**L and the center hook **70**C.

(96) Further, within a range in which the opening/closing pin **71***a* is moved in the direction of the arrow A**1** and the opening/closing pin **71***a* is located at the second locking portion **73***d* of the opening/closing guide hole **73**L, as shown with the broken line, the wire W is locked in a state in which the wire W is not allowed to move. Here, within the range in which the opening/closing pin **71***a* is located at the second locking portion **73***d* of the opening/closing guide hole **73**L, an operation of twisting the wire W with the binding unit **7** is performed, as described later.

(97) In a modified embodiment shown in FIG. **3**E, the opening/closing guide hole **73**L has a first

locking portion **73***b*, an unlocking portion **73***c*, and a second locking portion **73***d*. The unlocking portion **73***c* is configured to face, at a portion connected to the first lock portion **73***b*, the outer wall surface of the opening/closing guide hole **73**L with a dimension slightly greater than the diameter of the opening/closing pin **71***a*. In addition, the unlocking portion **73***c* is configured by an inclined surface inclined with respect to the outer wall surface, and is connected to the second lock portion **73***d*.

(98) In the modified embodiment shown in FIG. **3**E, the second side hook **70**L is configured to enable the opening/closing pin **71***a* to move along the inner wall surface of the opening/closing guide hole **73**L by an operation of the opening/closing pin **71***a* moving in the direction of the arrow A**1**, and to lock the wire W in a state in which the wire W is not allowed to move, within a range in which the opening/closing pin **71***a* is located at the first locking portion **73***b* of the opening/closing guide hole **73**L, as shown with the solid line.

(99) On the other hand, within a range in which the opening/closing pin **71***a* is moved in the direction of the arrow A**1** and the opening/closing pin **71***a* is located at the unlocking portion **73***c* of the opening/closing guide hole **73**L, as shown with the dashed-two dotted line, the opening/closing guide hole **73**L can be displaced up to a position denoted with the dashed-two dotted line, with respect to the opening/closing pin **71***a*, and the second side hook **70**L becomes movable in the direction of the arrow H**2** in which the second side hook **70**L is spaced apart from the center hook **70**C by such a predetermined amount that the wire W does not come off between the second side hook **70**L and the center hook **70**C. In addition, within a range in which the opening/closing pin **71***a* is located at the unlocking portion **73***c* of the opening/closing guide hole **73**L, as the opening/closing pin **71***a* comes closer to the second locking portion **73***d*, a movable amount in the direction in which the second side hook **70**L is spaced apart from the center hook **70**C becomes smaller.

(100) Further, within a range in which the opening/closing pin **71***a* is moved in the direction of the arrow A**1** and the opening/closing pin **71***a* is located at the second locking portion **73***d* of the opening/closing guide hole **73**L, as shown with the broken line, the wire W is locked in a state in which the wire W is not allowed to move.

(101) In a modified example shown in FIG. **3**F, the opening/closing guide hole **73**L has a first locking portion **73***b*, an unlocking portion **73***c*, and a second locking portion **73***d*. The unlocking portion **73***c* is configured to face, at a portion connected to the first lock portion **73***b*, the outer wall surface of the opening/closing guide hole **73**L with a dimension slightly greater than the diameter of the opening/closing pin **71***a*. In addition, the unlocking portion **73***c* is configured by an inclined surface inclined with respect to the outer wall surface, and is connected to the second lock portion **73***d*.

(102) The second locking portion **73***d* is configured by an inclined surface connected to the unlocking portion **73***c*. The second locking portion **73***d* is configured such that an interval between the inner wall surface and the outer wall surface of the opening/closing guide hole **73**L becomes smaller toward the front side of the opening/closing guide hole **73**L and the inner wall surface and the outer wall surface at an end portion on the front side of the opening/closing guide hole **73**L face each other with a dimension substantially equivalent to the diameter of the opening/closing pin **71***a*.

(103) In the modified embodiment shown in FIG. **3**F, the second side hook **70**L is configured to enable the opening/closing pin **71***a* to move along the inner wall surface of the opening/closing guide hole **73**L by an operation of the opening/closing pin **71***a* moving in the direction of the arrow A**1**, and to lock the wire W in a state in which the wire W is not allowed to move, within a range in which the opening/closing pin **71***a* is located at the first locking portion **73***b* of the opening/closing guide hole **73**L, as shown with the solid line.

(104) On the other hand, within a range in which the opening/closing pin **71***a* is moved in the direction of the arrow A**1** and the opening/closing pin **71***a* is located at the unlocking portion **73***c* of

the opening/closing guide hole **73**L, as shown with the dashed-two dotted line, the opening/closing guide hole **73**L can be displaced up to a position denoted with the dashed-two dotted line, with respect to the opening/closing pin **71***a*, and the second side hook **70**L becomes movable in the direction of the arrow H**2** in which the second side hook **70**L is spaced apart from the center hook **70**C by such a predetermined amount that the wire W does not come off between the second side hook **70**L and the center hook **70**C. In addition, within a range in which the opening/closing pin **71***a* is located at the unlocking portion **73***c* of the opening/closing guide hole **73**L, as the opening/closing pin **71***a* comes closer to the second locking portion **73***d*, a movable amount in the direction in which the second side hook **70**L is spaced apart from the center hook **70**C becomes smaller.
(105) Further, within a range in which the opening/closing pin **71***a* is moved in the direction of the arrow A**1** and the opening/closing pin **71***a* is located at the second locking portion **73***d* of the opening/closing guide hole **73**L, as shown with the broken line, the wire W is locked in a state in which the wire W is not allowed to move.
(106) Example of Embodiment of Cutting Unit
(107) FIGS. **4**A and **4**B are plan views showing an example of the cutting unit of the present embodiment, FIGS. **4**C to **4**E are perspective views showing the example of the cutting unit of the present embodiment, and FIGS. **4**F and **4**G are plan views showing modified embodiments of the cutting unit of the present embodiment. Next, an example of the cutting unit of the present embodiment will be described with reference to each drawing.
(108) The fixed blade part **60** is an example of the blade part, has a cylindrical shape serving as an axis of rotation of the movable blade part **61**, and is provided with an opening **60***a* penetrating in a radial direction of the cylindrical shape along the feeding path of the wire W. The opening **60***a* has a shape through which the wire W can pass. In the configuration in which the reinforcing bars S are bound with the two wires W, a cross-sectional shape of the opening **60***a* is a long hole shape along a direction in which the two wires W are aligned in parallel.
(109) Preferably, the opening **60***a* has, for example, a tapered shape in which opening areas on an introduction side and a discharge side of the opening **60***a* are widened with respect to the feeding of the wire W in the forward direction denoted with the arrow F. The fixed blade part **60** is provided on a downstream side of the wire guide **4** with respect to the feeding direction of the wire W that is conveyed in the forward direction.
(110) In the configuration in which the reinforcing bars S are bound with the two wires W, the fixed blade part **60** has a first butting portion **60***b* and a second butting portion **60***c* at an end portion of the opening **60***a* exposed on a circumferential surface on which the movable blade part **61** slides. The fixed blade part **60** is provided with a plurality of butting portions in a direction in which a plurality of wires W are aligned in parallel, and in the present example, is provided with the first butting portion **60***b*, which is one butting portion, and the second butting portion **60***c*, which is the other butting portion, along the direction in which the two wires W are aligned in parallel.
(111) The fixed blade part **60** is provided with the first butting portion **60***b* on a front side and the second butting portion **60***c* on an inner side, with respect to a moving direction of the movable blade part **61** denoted with an arrow D**1**. The fixed blade part **60** has a step portion **60***d* formed between the first butting portion **60***b* and the second butting portion **60***c* by recessing the second butting portion **60***c* with respect to the moving direction of the movable blade part **61** denoted with the arrow D**1**. A recessed amount is preferably about a half of the diameter of the wire W.
(112) The fixed blade part **60** has a regulation portion **60***e* configured to suppress the wire W butted against the first butting portion **60***b* from moving in a direction of the second butting portion **60***c*. The regulation portion **60***e* is a planar surface extending in a direction substantially orthogonal to the moving direction of the movable blade part **61** denoted with the arrow D**1**, and is provided between the first butting portion **60***b* and the step portion **60***d*.
(113) The movable blade part **61** is an example of the blade part, has a shape of sliding along the

circumferential surface of the fixed blade part **60**, and is configured to be in sliding contact with an open end of the opening **60***a* of the fixed blade part **60** by a rotating operation about the fixed blade part **60** serving as a fulcrum shaft.

(114) The cutting unit **6** has wall portions **62***a* and **62***b* configured to regulate introduction of foreign matters. The wall portions **62***a* and **62***b* are provided on upstream and downstream sides along a locus of the rotating operation of the movable blade part **61**, with respect to the opening **60***a* of the fixed blade part **60**. The wall portions **62***a* and **62***b* each have a shape following the locus of the rotating operation of the movable blade part **61** about the fixed blade part **60** serving as a fulcrum, and are configured to suppress foreign matters, such as wastes entering from an opening at a front end of the main body part **10** and shavings resulting from rubbing of the wire W and the reinforcing bar S, from entering the periphery of the movable blade part **61**. Thereby, it is possible to suppress a malfunction of the movable blade part **61** and an increase in load for rotating the movable blade part **61**.

(115) As for the cutting unit **6**, when the movable blade part **61** is rotated in the direction of the arrow D**1** from an initial position, the wire W having passed through the opening **60***a* of the fixed blade part **60** is pressed against the open end of the opening **60***a* by the movable blade part **61**. One wire W of the two wires W aligned in parallel is pressed against an end edge portion of the first butting portion **60***b* of the fixed blade part **60** by the operation of the movable blade part **61**, and the other wire W is introduced into the second butting portion **60***c* of the fixed blade part **60**. Thereby, a shearing force is applied to one wire W, and cutting of the one wire W is started prior to the other wire W.

(116) When the movable blade part **61** is rotated in the direction of the arrow D**1** to start cutting of the first wire W, which is one wire, and the first wire W is cut to a predetermined position, the second wire W, which is the other wire, is pressed against an end edge portion of the second butting portion **60***c* of the fixed blade part **60** by the operation of the movable blade part **61**.

(117) Thereby, cutting of the second wire W is started. Preferably, the shapes and positions of the first butting portion **60***b* and the second butting portion **60***c* are set so that, after starting the cutting of the first wire W, when the first wire W is cut in half or more in the radial direction, cutting of the second wire W is started. That is, a distance from the end edge portion of the first butting portion **60***b* to the end edge portion of the second butting portion **60***c* along the rotation direction of the movable blade part **61** denoted with the arrow D**1** is set to be a substantial half of the wire W in the radial direction.

(118) When the movable blade part **61** is further rotated in the direction of the arrow D**1**, the cutting of the one wire W for which cutting has been started first is completed. When the movable blade part **61** is further rotated to a cutting completion position in the direction of arrow D**1**, the cutting of the other wire W for which cutting has been started later is completed.

(119) The fixed blade part **60** has the regulation portion **60***e* formed between the first butting portion **60***b* and the second butting portion **60***c* and having a planar surface extending in a direction substantially orthogonal to the moving direction of the movable blade part **61** denoted with the arrow D**1**. Due to the planar surface, when the movable blade part **61** is moved in the direction of the arrow D**1**, it is possible to prevent an unintended force from acting on the wire W in the direction substantially orthogonal to the moving direction.

(120) Thereby, the wire W butted against the first butting portion **60***b* by the movable blade part **61** is suppressed from moving to the direction of the second butting portion **60***c*. In addition, the wire W is suppressed from moving in the direction of the second butting portion **60***c*, so that wear of the step portion **60***d* is suppressed and a difference in distance from the end edge portion of the first butting portion **60***b* to the end edge portion of the second butting portion **60***c* along the rotation direction of the movable blade part **61** denoted with the arrow D**1** is suppressed from decreasing. Therefore, it is possible to secure a phase difference of timings at which the cuttings of the two wires W are started, and to suppress an increase in load, which is caused when the cuttings of the

two wires W are started at substantially the same time.
(121) Note that, the regulation portion **60***e* may be configured by providing the planar surface, which extends in the direction substantially orthogonal to the moving direction of the movable blade part **61** denoted with the arrow D**1**, at a part between the first butting portion **60***b* and the step portion **60***d*. In addition, the regulation portion **60***e* may be configured by an inclined surface or a curved surface where the step portion **60***d* protrudes from the first butting portion **60***b* toward the second butting portion **60***c* along a direction (arrow D**2**) opposite to the moving direction of the movable blade part **61** denoted with the arrow D**1**.
(122) Further, as shown in FIG. **4**F, the regulation portion **60***e* may be configured by a convex portion protruding from the first butting portion **60***b* and the second butting portion **60***c* along the direction (arrow D**2**) opposite to the moving direction of the movable blade part **61** denoted with the arrow D**1**, between the first butting portion **60***b* and the second butting portion **60***c*. Thereby, the first butting portion **60***b* becomes a concave shape, so that the wire W butted against the first butting portion **60***b* by the movable blade part **61** is suppressed from moving in the direction of the second butting portion **60***c*.
(123) Further, as shown in FIG. **4**G, the regulation portion **60***e* may be formed into a shape of partitioning the first butting portion **60***b* and the second butting portion **60***c* therebetween. Thereby, the first butting portion **60***b* and the second butting portion **60***c* are made independent, so that the wire W butted against the first butting portion **60***b* by the movable blade part **61** is suppressed from moving in the direction of the second butting portion **60***c*.
(124) Example of Embodiment of Transmission Unit
(125) Next, an example of the transmission unit **9** of the present embodiment will be described with reference to each drawing. The transmission unit **9** is supported so that the cam **90** can rotate about a shaft **90***a* as a fulcrum. The shaft **90***a* is attached to a frame **10***a* attached to an interior of the main body part **10**. The frame **10***a* has a guide portion **10***b* configured to regulate a moving direction of a link **91**. The guide portion **10***b* is configured by a long hole penetrating through the plate-shaped frame **10***a*.
(126) The cam **90** is an example of the displacement member, and has a cam groove **92** whose length from the shaft **90***a* is displaced. The cam groove **92** extends in radial and circumferential directions of the cam **90** about the shaft **90***a*, and intersects the guide portion **10***b* of the frame **10***a*. The cam groove **92** penetrates through the plate-shaped cam **90**, so that an intersection of the cam groove **90** and the guide portion **10***b* communicates.
(127) The cam **90** is configured such that a rotating operation about the shaft **90***a* as a fulcrum changes a portion of the cam groove **92** intersecting the guide portion **10***b*, thereby changing a length from the shaft **90***a* to the intersection of the cam groove **92** and the guide portion **10***b*.
(128) For the cam **90**, ranges in which an amount of change in length between the shaft **90***a* and the cam groove **92** by the rotating operation about the shaft **90***a* as a fulcrum is large and small for the same amount of rotation of the cam **90** are set. In the present example, a first range **92***a* in which the amount of change in length between the shaft **90***a* and the cam groove **92** is the largest, a second range **92***b* in which the amount of change in length between the shaft **90***a* and the cam groove **92** is smaller than the first range **92***a*, and a third range **92***c* in which there is little amount of change in length between the shaft **90***a* and the cam groove **92** are provided.
(129) The cam **90** is configured such that, while the first range **92***a* of the cam groove **92** intersects the guide portion **10***b* by the rotating operation in the direction of the arrow C**1** about the shaft **90***a* as a fulcrum, the length from the shaft **90***a* to the intersection of the cam groove **92** and the guide portion **10***b* is shorter and the amount of change in length between the shaft **90***a* and the cam groove **92** becomes larger, as compared with a case where the second range **92***b* intersects the guide portion **10***b*.
(130) In addition, the cam **90** is configured such that, while the second range **92***b* of the cam groove **92** intersects the guide portion **10***b* by the rotating operation in the direction of the arrow C**1** about

the shaft **90***a* as a fulcrum, the length from the shaft **90***a* to the intersection of the cam groove **92** and the guide portion **10***b* is longer and the amount of change in length between the shaft **90***a* and the cam groove **92** becomes smaller, as compared with the case where the first range **92***a* intersects the guide portion **10***b*.

(131) Further, the cam **90** is configured such that, while the third range **92***c* of the cam groove **92** intersects the guide portion **10***b* by the rotating operation in the direction of the arrow C**1** about the shaft **90***a* as a fulcrum, the length from the shaft **90***a* to the intersection of the cam groove **92** and the guide portion **10***b* is substantially equivalent and the amount of change in length between the shaft **90***a* and the cam groove **92** is further smaller and substantially constant, as compared with the case where the second range **92***b* intersects the guide portion **10***b*.

(132) The cam **90** has an engaged portion **93** to which movement of the sleeve **71** is transmitted via the moving member **75**. The engaged portion **93** is provided on an opposite side to the cam groove **92** with the shaft **90***a* interposed therebetween, and is arranged on a locus of the engaging portion **75***a* by the movement of the moving member **75** in conjunction with the movement of the sleeve **71** in the front and rear direction denoted with the arrows A**1** and A**2**. The engaged portion **93** is engaged with the engaging portion **75***a* of the moving member **75** by an operation in which the sleeve **71** is moved in the forward direction denoted with the arrow A**1**.

(133) The cam **90** is urged by a spring **94** in the direction of the arrow C**2** in which the first range **92***a* of the cam groove **92** intersects the guide portion **10***b* by the rotating operation about the shaft **90***a* as a fulcrum. The spring **94** is configured by, for example, a torsion coil spring attached to the shaft **90***a*. Note that, the rotation direction of the cam **90** denoted with the arrow C**2** corresponds to a direction in which the movable blade part **61** connected by the link **91** returns from the cutting completion position to the initial position. In consideration of a case in which the cam **90** cannot rotate in the direction of the arrow C**2** with the force of the spring **94** by the operation of the movable blade part **61** returning from the cutting completion position to the initial position, the moving member **75** is provided with a pressing convex portion **76** and the cam **90** is provided with a pressed convex portion **96**. When the moving member **75** is moved in the direction of the arrow A**1** direction and the cam **90** is rotated until the movable blade part **61** is rotated to the cutting completion position, the pressing convex portion **76** and the pressed convex portion **96** face. By the operation of the sleeve **71** moving in the direction of the arrow A**2**, the pressing convex portion **76** pushes the pressed convex portion **96**, so that the cam **90** can be forced to start rotating in the direction of the arrow C**2**.

(134) The link **91** is an example of the transmission member, and has an end portion in the forward direction denoted with the arrow A**1** connected to the movable blade part **61**, and an end portion in the backward direction denoted with the arrow A**2** connected to the cam **90**.

(135) The link **91** has a shaft portion **91***a* configured to enter the cam groove **92** of the cam **90** and the guide portion **10***b* of the frame **10***a*. The shaft portion **91***a* is configured by a rotary body **91***a***1** configured to enter the cam groove **92**, and a shaft **91***a***2** configured to rotatably support the rotary body **91***a***1** and to be non-rotatable with respect to the link **91** that enters the guide portion **10***b*, and is inserted into the cam groove **92** and the guide portion **10***b* at the intersection of the cam groove **92** and the guide portion **10***b*. The shaft portion **91***a* is configured to move along the cam groove **92** and the guide portion **10***b* by the rotating operation of the cam **90** about the shaft **90***a* as a fulcrum. Here, by the rotating operation of the cam **90** about the shaft **90***a* as a fulcrum, a force that is applied in a circumferential direction of the rotary body **91***a***1** as the cam groove **92** and the rotary body **91***a***1** are slid and a force that is applied in a circumferential direction of the shaft **91***a***2** as the guide portion **10***b* and the shaft **91***a***2** are slid become forces in opposite directions. Therefore, in the shaft portion **91***a*, the rotary body **91***a***1** and the shaft **91***a***2** are configured as separate components. Note that, the shaft portion **91***a* may have a first rotary body configured to enter the cam groove **92**, a second rotary body configured to enter the guide portion **10***b*, and a shaft configured to rotatably support the first rotary body and the second rotary body.

(136) When the sleeve **71** is moved in the forward direction denoted with the arrow A**1**, the moving member **75** is moved in the forward direction denoted with the arrow A**1** in conjunction with the sleeve **71**. The moving member **75** is configured such that the engaging portion **75***a* is engaged with the engaged portion **93** of the cam **90** by the moving operation in the forward direction denoted with the arrow A**1**.
(137) When the moving member **75** is further moved in the forward direction denoted with the arrow A**1**, the engaged portion **93** is pushed forward, so that the cam **90** is rotated in the direction of the arrow C**1** about the shaft **90***a* as a fulcrum. When the cam **90** is rotated in the direction of the arrow C**1**, a portion of the cam groove **92** intersecting the guide portion **10***b* changes, and the length from the shaft **90***a* to the intersection of the cam groove **92** and the guide portion **10***b* changes in an increasing direction.
(138) Thereby, when the cam **90** is rotated in the direction of the arrow C**1** and the shaft portion **91***a* of the link **91** is moved along the cam groove **92** and the guide portion **10***b*, the shaft portion **91***a* is moved in a direction away from the shaft **90***a* of the cam **90**.
(139) The transmission unit **9** is configured such that, when the shaft portion **91***a* of the link **91** is moved in the direction away from the shaft **90***a* of the cam **90**, the rotating operation of the cam **90** is converted into movement along the extension direction of the link **91**.
(140) Thereby, the rotating operation of the cam **90** is transmitted to the movable blade part **61** via the link **91**, so that the movable blade part **61** is rotated in the direction of the arrow D**1**. Therefore, the moving operation of the sleeve **71** in the forward direction rotates the movable blade part **61** in a predetermined direction to cut the wire W.
(141) A period during which the first range **92***a* of the cam groove **92** intersects the guide portion **10***b* corresponds to a period after the movable blade part **61** of the cutting unit **6** starts rotation until the cutting of the first wire W is started. The period until the cutting of the first wire W is started corresponds to a region in which a load is low.
(142) In addition, a period during which the second range **92***b* of the cam groove **92** intersects the guide portion **10***b* corresponds to a period after the movable blade part **61** of the cutting unit **6** rotates and the cutting of the first wire W is started until the cutting of the second wire W ends. The period after the cutting of the first wire W is started until the cutting of the second wire W ends corresponds to a region in which a load is high. Further, a period during which the third range **92***c* of the cam groove **92** intersects the guide portion **10***b* corresponds to a period during which the cutting of the second wire W ends and the rotation of the movable blade part **61** stops. In this way, with respect to the amount of movement of the moving member **75**, it is not necessary to rotate the cutter having completed the wire cutting operation more than necessary.
(143) Note that, in the above embodiment, the cam **90** has such a configuration that the length from the intersection of the cam groove **92**, which is a first connection portion connected to the link **91**, and the guide portion **10***b* to the shaft **90***a* is switched by the rotating operation about the shaft **90***a* as a fulcrum due to the shape of the cam groove **92**.
(144) Thereby, the cam **90** makes it possible to switch the amount of rotation (amount of movement) of the movable blade part **61** and the force that can be generated by the movable blade part **61**, within the rotating range (moving range) of the movable blade part **61**.
(145) On the other hand, the cam **90** may be configured such that a length from the engaged portion **93**, which is a second connection portion connected to the sleeve **71**, to the shaft **90***a* is switched by the rotating operation about the shaft **90***a* as a fulcrum.
(146) Example of Embodiment of Decelerator
(147) FIG. **5**A is a side cross-sectional view showing an example of the decelerator of the present embodiment, FIG. **5**B is a perspective view showing the example of the decelerator of the present embodiment, FIG. **5**C is a side cross-sectional view of main parts showing a modified embodiment of the decelerator of the present embodiment, and FIG. **5**D is a perspective view showing the modified embodiment of the decelerator of the present embodiment. Next, an example of the

decelerator of the present embodiment will be described with reference to each drawing.
(148) The decelerator **81** is configured by a planet gear in which an input shaft and an output shaft are coaxially arrayed, and includes a first sun gear **82***a* attached to a shaft **80***a* of a motor **80** serving as an input shaft, a first planetary gear **83***a* in mesh with the first sun gear **82***a* and a first planet cage **84***a* configured to support the first planetary gear **83***a*.
(149) In addition, the decelerator **81** includes a second sun gear **82***b* provided to the first planet cage **84***a*, a second planetary gear **83***b* in mesh with the second sun gear **82***b*, and a second planet cage **84***b* configured to support the second planetary gear **83***b*.
(150) Further, the decelerator **81** includes an internal gear **85** in mesh with the first planetary gear **83***a* and the second planetary gear **83***b*.
(151) As for the decelerator **81**, the internal gear **85** is fixed to the main body part **10**. In addition, as for the decelerator **81**, the first planet cage **84***a* and the second planet cage **84***b* are arranged coaxially with the shaft **80***a* of the motor **80**. Further, as for the decelerator **81**, the second planet cage **84***b* is connected to the rotary shaft **72**, and configures an output shaft.
(152) As for the decelerator **81**, a front side portion **84***f* that is one side along an axis direction of the second planet cage **84***b* protrudes from the internal gear **85**. As for the second planet cage **84***b*, the front side portion **84***f* protruding from the internal gear **85** is rotatably supported by the main body part **10** via a bearing **86**.
(153) In addition, as for the second planet cage **84***b*, a rear side portion **84***r* that is the other side along the axis direction is located inside the internal gear **85**, and the rear side portion **84***r* is supported to the internal gear **85** by a support member **87**. Since the internal gear **85** is fixed to the main body part **10**, the rear side portion **84***r* of the second planet cage **84***b* is supported by the main body part **10** via the support member **87** configuring a sliding bearing and the internal gear **85**. Note that, the support member **87** may be configured by a bearing.
(154) Further, the decelerator **81** includes a gear holder **88** between the first planet cage **84***a* and the second planetary gear **83***b*. The gear holder **88** is configured by a disk-shaped member having a hole perforated at a center into which the second sun gear **82***b* is inserted, and is inserted between the first planet cage **84***a* and the second planetary gear **83***b* outside the second sun gear **82***b*, thereby securing a gap between the first planet cage **84***a* and the second planetary gear **83***b*.
(155) Thereby, the second planet cage **84***b* is supported at the front side portion **84***f* and the rear side portion **84***r* along the axis direction by the main body part **10**. Therefore, with a simple configuration, the second planet cage **84***b* is suppressed from being inclined with respect to the axis direction, and changes in meshes between the sun gear and the planetary gear and between the planetary gear and the internal gear, and interferences between gears aligned in parallel in the axis direction, between a gear and a planet cage, and the like are suppressed.
(156) Note that, like the decelerator **81** of a modified embodiment shown in FIGS. **5**C and **5**D, the gear holder **88***a* may be provided integrally with the first planet cage **84***a*. The gear holder **88***a* is configured such that a disk-shaped member having a hole perforated at a center into which the second sun gear **82***b* is inserted is provided integrally with the first planet cage **84***a* outside the second sun gear **82***b*. Thereby, the gear holder **88***a* is inserted between the first planet cage **84***a* and the second planetary gear **83***b* outside the second sun gear **82***b*, thereby securing a gap between the first planet cage **84***a* and the second planetary gear **83***b*.
(157) Example of Embodiment of Curl Forming Unit
(158) FIGS. **6**A to **6**D are plan views showing an example of the curl forming unit of the present embodiment. Next, an example of the curl forming unit of the present embodiment will be described with reference to each drawing.
(159) The curl forming unit **5** includes a guide groove **52** configuring a feeding path of the wire W in the curl forming unit **5**, and a first guide member **53***a* and a second guide member **53***b*, which are configured to curl the wire W in cooperation with the guide groove **52**.
(160) The first guide member **53***a* is provided on an introduction part side of the curl guide **50** for

the wire W that is fed in the forward direction by the wire feeding unit **3**, and is arranged on a radially inner side of the loop Ru formed by the wire W with respect to the feeding path of the wire W by the guide groove **52**. The first guide member **53***a* is configured to regulate the feeding path of the wire W so that the wire W fed along the guide groove **52** does not enter the radially inner side of the loop Ru formed by the wire W.

(161) The second guide member **53***b* is provided on a discharge part side of the curl guide **50** for the wire W that is fed in the forward direction by the wire feeding unit **3**, and is arranged on a radially outer side of the loop Ru formed by the wire W with respect to the feeding path of the wire W by the guide groove **52**.

(162) The curl forming unit **5** includes a retraction mechanism **54** configured to retract the first guide member **53***a* from the feeding path of the wire W. The retraction mechanism **54** is attached to a frame **55** for fixing the curl guide **50** to the main body part **10** so as to be rotatable about a shaft **54***a* as a fulcrum, and is configured to be displaced in directions in which the first guide member **53***a* protrudes and retracts with respect to the feeding path of the wire W.

(163) The retraction mechanism **54** is urged by an urging member **56** such as a spring, in the direction in which the first guide member **53***a* protrudes to the feeding path of the wire W.

(164) In addition, the retraction mechanism **54** includes an induction part **57** configured to displace the retraction mechanism **54** in the direction in which the first guide member **53***a* retracts with respect to the feeding path of the wire W. The induction part **57** is configured by an inclined surface configured, in an operation of winding the wire W on the reinforcing bars S, to be pushed by the wire W, thereby generating a force for displacing the retraction mechanism **54** in the direction in which the first guide member **53***a* retracts with respect to the feeding path of the wire W.

(165) In addition, the retraction mechanism **54** includes a wire guide part **58** configuring a part of the guide groove **52**. When the retraction mechanism **54** is moved in the direction in which the first guide member **53***a* protrudes with respect to the feeding path of the wire W, the wire guide part **58** protrudes to the feeding path of the wire W, and configures a part of the guide groove **52**. In addition, when the retraction mechanism **54** is moved in the direction in which the first guide member **53***a* retracts with respect to the feeding path of the wire W, the wire guide part **58** protrudes to the feeding path of the wire W, and closes a path along which the wire W is exposed to an outside of the guide groove **52**.

(166) The curl forming unit **5** includes a feeding regulation part **59** against which a tip end of the wire W is butted, on the feeding path of the wire W that is curled by the curl guide **50** and guided to the binding unit **7** by the induction guide **51**.

(167) The retraction mechanism **54** includes an opening/closing regulation portion **54***b* configured to engage with the moving member **75** configured to move in conjunction with the sleeve **71** and to be in contact with an opening/closing regulation member **55***a* configured to operate in conjunction with the moving member **75**. The opening/closing regulation portion **54***b* comes in contact with the opening/closing regulation member **55***a* in a state in which the retraction mechanism **54** has moved in the direction in which the first guide member **53***a* protrudes to the feeding path of the wire W, so that the rotation of the retraction mechanism **54** about the shaft **54***a* as a fulcrum is regulated.

(168) In addition, when the opening/closing regulation member **55***a* is moved in conjunction with the operation of the binding unit **7** for locking the wire W with the locking member **70**, and an opening portion **55***b* of the opening/closing regulation member **55***a* is moved to a position where it faces the opening/closing regulation portion **54***b* of the retraction mechanism **54**, the opening/closing regulation portion **54***b* enters the opening portion **55***b*, so that the regulation of rotation of the retraction mechanism **54** about the shaft **54***a* as a fulcrum is released. Thereby, the retraction mechanism **54** can be moved by the rotating operation about the shaft **54***a* as a fulcrum, in the direction in which the first guide member **53***a* retracts with respect to the feeding path of the wire W.

(169) <Example of Operation of Reinforcing Bar Binding Machine of Present Embodiment>

(170) Subsequently, an operation of binding the reinforcing bars S with the wire W by the reinforcing bar binding machine **1**A of the present embodiment will be described with reference to each drawing.
(171) The reinforcing bar binding machine **1**A is in a standby state where the wire W is sandwiched between the pair of feeding gears **30** and the tip end of the wire W is located between a sandwiched position by the feeding gears **30** and the fixed blade part **60** of the cutting unit **6**. Also, when the reinforcing bar binding machine **1**A is in the standby state, the sleeve **71** and the first side hook **70**R, the second side hook **70**L and the center hook **70**C attached to the sleeve **71** are moved in the rear direction denoted with the arrow A**2**, and as shown in FIG. **3**A, the first side hook **70**R is opened with respect to the center hook **70**C, and the second side hook **70**L is opened with respect to the center hook **70**C.
(172) When the reinforcing bars S are inserted between the curl guide **50** and the induction guide **51** of the curl forming unit **5** and a trigger **12** is operated, the feeding motor (not shown) is driven in the forward rotation direction, so that the wire W is fed in the forward direction denoted with the arrow F by the wire feeding unit **3**A.
(173) In a configuration where a plurality of, for example, two wires W are fed, the two wires W are fed aligned in parallel along an axis direction of the loop Ru, which is formed by the wires W, by the wire guide **4**.
(174) The wire W fed in the forward direction passes between the center hook **70**C and the first side hook **70**R, and is then fed to the curl guide **50** of the curl forming unit **5**. The wire W passes through the curl guide **50** and is thus curled to be wound around the reinforcing bars S.
(175) The wire W curled by the curl guide **50** is guided to the induction guide **51** and is further fed in the forward direction by the wire feeding unit **3**A, so that the wire is guided between the center hook **70**C and the second side hook **70**L by the induction guide **51**. Then, the wire W is fed until the tip end is butted against the feeding regulation part **59**. When the wire W is fed to a position at which the tip end is butted against the feeding regulation part **59**, the drive of the feeding motor (not shown) is stopped.
(176) After stopping the feeding of the wire W in the forward direction, the motor **80** is driven in the forward rotation direction. In the first operation area where the wire W is locked by the locking member **70**, the rotation regulation blade **74***a* is locked, so that the rotation of the sleeve **71** in conjunction with the rotation of the rotary shaft **72** is regulated. Thereby, the rotation of the motor **80** is converted into linear movement, so that the sleeve **71** is moved in the forward direction denoted with the arrow A**1**.
(177) When the sleeve **71** is moved in the forward direction denoted with the arrow A**1**, the first side hook **70**R and the second side hook **70**L of the locking member **70** are moved toward the center hook **70**C by the rotating operations about the shaft **71***b* as a fulcrum, due to the locus of the opening/closing pin **71***a* and the shape of the opening/closing guide holes **73**R and **73**L.
(178) That is, when the sleeve **71** is moved in the forward direction denoted with the arrow A**1**, the inner wall surface of the first side hook **70**R with respect to the direction in which the first side hook **70**R is closed is pushed by the opening/closing pin **71***a*, in the opening/closing portion **73***a* formed in the opening/closing guide hole **73**R. Thereby, the first side hook **70**R is rotated about the shaft **71***b* as a fulcrum and is moved toward the center hook **70**C.
(179) In addition, when the sleeve **71** is moved in the forward direction denoted with the arrow A**1**, the inner wall surface of the second side hook **70**L with respect to the direction in which the second side hook **70**L is closed is pushed by the opening/closing pin **71***a*, in the opening/closing portion **73***a* formed in the opening/closing guide hole **73**L. Thereby, the second side hook **70**L is rotated about the shaft **71***b* as a fulcrum and is moved toward the center hook **70**C.
(180) Thereby, the first side hook **70**R and the second side hook **70**L are closed with respect to the center hook **70**C.
(181) When the first side hook **70**R is closed with respect to the center hook **70**C, the wire W

sandwiched between the first side hook **70**R and the center hook **70**C is locked in such a manner that the wire can move between the first side hook **70**R and the center hook **70**C.
(182) On the other hand, when the second side hook **70**L is closed with respect to the center hook **70**C, the wire W sandwiched between the second side hook **70**L and the center hook **70**C is locked in such a manner that the wire cannot come off between the second side hook **70**L and the center hook **70**C, within the range in which the opening/closing pin **71***a* is located at the locking portion **73***b* of the opening/closing guide hole **73**L, as shown in FIG. **3**B.
(183) After advancing the sleeve **71** to a position, at which the opening/closing pin **71***a* is located at the locking portion **73***b* of the opening/closing guide hole **63**L and the wire W is locked, by the closing operation of the first side hook **70**R and the second side hook **70**L, the rotation of the motor **80** is temporarily stopped and the feeding motor (not shown) is driven in the reverse rotation direction.
(184) Thereby, the pair of feeding gears **30** is reversely rotated and the wire W sandwiched between the pair of feeding gears **30** is fed in the reverse direction denoted with the arrow R. Since the tip end side of the wire W is locked in such a manner that the wire does not come off between the second side hook **70**L and the center hook **70**C, the wire W is wound on the reinforcing bars S by the operation of feeding the wire W in the reverse direction.
(185) In addition, in the operation of winding the wire W on the reinforcing bars S, the induction part **57** of the retraction mechanism **54** is pushed by the wire W, so that the first guide member **53***a* retracts with respect to the feeding path of the wire W.
(186) After the wire W is wound on the reinforcing bars S and the drive of the feeding motor (not shown) in the reverse rotation direction is stopped, the motor **80** is driven in the forward rotation direction, so that the sleeve **71** is further moved in the forward direction denoted with the arrow A**1**.
(187) FIGS. **7**A to **7**G are operation explanatory diagrams showing an example of the operations of the binding unit, the transmission unit and the cutting unit according to the present embodiment. As shown in FIG. **7**A, when the sleeve **71** is moved in the forward direction denoted with the arrow A**1**, the moving member **75** is moved in the forward direction denoted with the arrow A**1** in conjunction with the sleeve **71**.
(188) As shown in FIG. **7**B, the engaging portion **75***a* is engaged with the engaged portion **93** of the cam **90** by the operation of the moving member **75** moving in the forward direction denoted with the arrow A**1**. A region from when the sleeve **71** is moved in the forward direction denoted with the arrow A**1** until the engaging portion **75***a* of the moving member **75** is engaged with the engaged portion **93** of the cam **90** is referred to as an idle running region.
(189) When the moving member **75** is further moved in the forward direction denoted with the arrow A**1**, the engaged portion **93** is pushed forward, so that the cam **90** is rotated in the direction of the arrow C**1** about the shaft **90***a* as a fulcrum. When the cam **90** is rotated in the direction of the arrow C**1**, a portion of the cam groove **92** intersecting the guide portion **10***b* changes, and the length from the shaft **90***a* of the cam **90** to the intersection of the cam groove **92** and the guide portion **10***b* changes in an increasing direction.
(190) As for the link **91**, the shaft portion **91***a* is inserted into the cam groove **92** and the guide portion **10***b* at the intersection of the cam groove **92** and the guide portion **10***b*, and the rotating operation of the cam **90** about the shaft **90***a* as a fulcrum moves the shaft portion **91***a* along the cam groove **92** and the guide portion **10***b*.
(191) Thereby, when the cam **90** is rotated in the direction of the arrow C**1** and the length from the shaft **90***a* of the cam **90** to the intersection of the cam groove **92** and the guide portion **10***b* changes in an increasing direction, the shaft portion **91***a* of the link **91** is moved along the cam groove **92** and the guide portion **10***b*, so that the shaft portion **91***a* is moved in the direction away from the shaft **90***a* of the cam **90**.
(192) As for the transmission unit **9**, when the shaft portion **91***a* of the link **91** is moved in the direction away from the shaft **90***a* of the cam **90**, the rotating operation of the cam **90** is converted

into movement along the extension direction of the link **91**.
(193) Thereby, the rotating operation of the cam **90** is transmitted to the movable blade part **61** via the link **91**, so that the movable blade part **61** is rotated in the direction of the arrow D**1**.
(194) When the movable blade part **61** is rotated in the direction of the arrow D**1**, one wire W of the two wires W aligned in parallel is pressed against the end edge portion of the first butting portion **60***b* of the fixed blade part **60** by the operation of the movable blade part **61**, and the other wire W enters the second butting portion **60***c* of the fixed blade part **60**, so that the cutting of the one wire W is started prior to the other wire W.
(195) A region from when the cam **90** is rotated in the direction of the arrow C**1** about the shaft **90***a* as a fulcrum, so that the movable blade **61** is rotated in the direction of the arrow D**1** until the cutting of the first wire W by the movable blade part **61** is started, as shown in FIG. **7**C, is referred to as an idling region. The idle running region and the idling region are regions in which a load that is applied to the movable blade part **61** is low.
(196) In the idling region, the first range **92***a* of the cam groove **92** intersects the guide portion **10***b*. While the first range **92***a* of the cam groove **92** intersects the guide portion **10***b*, the length from the shaft **90***a* to the intersection of the cam groove **92** and the guide portion **10***b* is shorter and the amount of change in length between the shaft **90***a* and the cam groove **92** becomes larger, as compared with the case where the second range **92***b* intersects the guide portion **10***b*.
(197) Thereby, the amount of rotation of the movable blade part **61** becomes relatively large with respect to the amount of movement of the sleeve **71** that rotates the cam **90**. On the other hand, in the idling region, since the cutting of the wire W has not been started, there is no wire cutting load that is applied to the movable blade part **61**, so that an increase in load that is applied to the cam **90** connected to the movable blade part **61** via the link **91** is suppressed.
(198) Since the cam **90** is connected to the sleeve **71** via the moving member **75**, the increase in load that is applied to the cam **90** is suppressed, so that an increase in load that is applied to the rotary shaft **72** that moves the sleeve **71** and to the motor **80** connected to the rotary shaft **72** via the decelerator **81** is suppressed.
(199) Therefore, in the region in which the load is low until the cutting of the first wire W is started, a time consumed to rotate the movable blade part **61** to a position where the cutting of the wire W is started can be shortened by relatively increasing the amount of rotation of the movable blade part **61**.
(200) When the moving member **75** is moved in the forward direction denoted with the arrow A**1** to the position where the movable blade part **61** starts cutting of the first wire W, the cam **90** rotates about the shaft **90***a* as a fulcrum, as shown in FIG. **7**D, so that the second range **92***b* of the cam groove **92** intersects the guide portion **10***b*.
(201) While the second range **92***b* of the cam groove **92** intersects the guide portion **10***b*, the length from the shaft **90***a* of the cam **90** to the intersection of the cam groove **92** and the guide portion **10***b* changes in an increasing direction, and the shaft portion **91***a* of the link **91** is moved along the cam groove **92** and the guide portion **10***b*, so that the shaft portion **91***a* is moved in the direction away from the shaft **90***a* of the cam **90**.
(202) Thereby, the moving member **75** is further moved in the forward direction denoted with the arrow A**1** to rotate the cam **90** in the direction of the arrow C**1**, and the rotating operation of the cam **90** is transmitted to the movable blade part **61** via the link **91**, so that the movable blade part **61** is further rotated in the direction of the arrow D**1** to start cutting of the first wire W.
(203) After the movable blade part **61** is rotated in the direction of the arrow D**1** to start cutting of the first wire W, which is one wire, when the first wire W is cut to a predetermined position, the second wire W, which is the other wire, is pressed against the end edge portion of the second butting portion **60***c* of the fixed blade part **60** by the operation of the movable blade part **61**.
(204) Thereby, cutting of the second wire W is started. In the present example, after starting the cutting of the first wire W, when the first wire W is cut in half or more in the radial direction, the

cutting of the second wire W is started.
(205) As described above, while the cutting of the first wire W is started and the second range **92***b* of the cam groove **92** intersects the guide portion **10***b*, the length from the shaft **90***a* to the intersection of the cam groove **92** and the guide portion **10***b* is longer and the amount of change in length between the shaft **90***a* and the cam groove **92** becomes smaller, as compared with the case where the first range **92***a* intersects the guide portion **10***b*.
(206) Thereby, the amount of rotation of the movable blade part **61** becomes relatively small with respect to the amount of movement of the sleeve **71**. On the other hand, the force that can be generated by the movable blade part **61** by operating the movable blade part **61** with the cam **90** via the link **91** increases.
(207) When the cutting of the first wire W is started, the load that is applied to the movable blade part **61** increases. On the other hand, the force that can be generated by the movable blade part **61** increases, so that the load that is applied to the movable blade part **61** is canceled and the increase in load that is applied to the cam **90** connected to the movable blade part **61** via the link **91** is suppressed.
(208) The increase in load that is applied to the cam **90** is suppressed, so that an increase in load that is applied to the rotary shaft **72** that moves the sleeve **71** and to the motor **80** connected to the rotary shaft **72** via the decelerator **81** is suppressed.
(209) When the movable blade part **61** is rotated in the direction of the arrow D**1** and the moving member **75** is moved in the forward direction denoted with the arrow A**1** from the position where the cutting of the first wire W is started to the position where the cutting of the second wire W is started, the cam **90** is rotated about the shaft **90***a* as a fulcrum, as shown in FIG. **7**E, so that the second range **92***b* of the cam groove **92** intersects the guide portion **10***b*.
(210) When the movable blade part **61** is further rotated in the direction of the arrow D**1**, the cutting of the one wire W for which cutting has been started first is completed. When the movable blade part **61** is further rotated in the direction of the arrow D**1**, the cutting of the other wire W for which cutting has been started later is completed.
(211) When the movable blade part **61** is rotated in the direction of the arrow D**1** and the moving member **75** is moved in the forward direction denoted with the arrow A**1** from a position where the cutting of the second wire W is started to a position where the cutting of the second wire W ends, as described above, the cam **90** is rotated about the shaft **90***a* as a fulcrum, as shown in FIG. **7**F, so that the second range **92***b* of the cam groove **92** intersects the guide portion **10***b*.
(212) When the cutting of the second wire W is started, the load that is applied to the movable blade part **61** further increases. On the other hand, the force that can be generated by the movable blade part **61** increases, so that the load that is applied to the movable blade part **61** is canceled and the increase in load that is applied to the cam **90** connected to the movable blade part **61** via the link **91** is suppressed.
(213) The increase in load that is applied to the cam **90** is suppressed, so that an increase in load that is applied to the rotary shaft **72** that moves the sleeve **71** and to the motor **80** connected to the rotary shaft **72** via the decelerator **81** is suppressed.
(214) Therefore, in a region in which the load is high from when the cutting of the first wire W is started until the cutting of the second wire W ends, the increase in load that is applied to the motor **80** can be suppressed by increasing the force that can be generated by the movable blade part **61**. In addition, in the region in which the load is high, the amount of rotation of the movable blade part **61** becomes relatively small, but in the region in which the load is low, the time consumed until the cutting of the wire W ends can be suppressed from lengthening by relatively increasing the amount of rotation of the movable blade part **61**.
(215) When the moving member **75** is moved in the forward direction denoted with the arrow A**1** to the position where the movable blade part **61** ends the cutting of the second wire W, the cam **90** is rotated about the shaft **90***a* as a fulcrum, as shown in FIG. **7**G, so that the third range **92***c* of the

cam groove **92** intersects the guide portion **10***b*.
(216) While the third range **92***c* of the cam groove **92** intersects the guide portion **10***b*, the length from the shaft **90***a* to the intersection of the cam groove **92** and the guide portion **10***b* is substantially equivalent and the amount of change in length between the shaft **90***a* and the cam groove **92** is further smaller and becomes substantially constant, as compared with the case where the second range **92***b* intersects the guide portion **10***b*.
(217) Thereby, the relative amount of rotation of the movable blade part **61** becomes smaller with respect to the amount of movement of the sleeve **71**. When the cutting of the wire W ends, it is not necessary to rotate the movable blade part **61**. On the other hand, after the cutting of the wire W, in order to bend the wire W, the sleeve **71** needs to be moved in the forward direction denoted with the arrow A**1**.
(218) Therefore, while the third range **92***c* of the cam groove **92** intersects the guide portion **10***b*, the amount of rotation of the movable blade part **61** is reduced with respect to the amount of movement of the sleeve **71**, and the increase in load due to the rotation of the movable blade part **61** after the cutting of the wire W is suppressed, so that the increase in load that is applied to the cam **90** connected to the movable blade part **61** via the link **91** is suppressed.
(219) Therefore, in the region from when the cutting of the second wire W ends until the movement of the sleeve **71** is stopped, the increase in load that is applied to the cam **90** due to the rotation of the movable blade part **61** is suppressed, so that the increase in load that is applied to the rotary shaft **72** that moves the sleeve **71** and to the motor **80** connected to the rotary shaft **72** via the decelerator **81** can be suppressed.
(220) Note that, the amount of movement of the sleeve **71** per rotation of the rotary shaft **72** is defined by a lead angle of the feeding screw **72***a*. Therefore, the lead angle of the feeding screw **72***a* is increased with respect to the reinforcing bar binding machine of the related art. On the other hand, in the region in which the load that is applied to the movable blade part **61** is high, the amount of rotation of the movable blade part **61** becomes relatively small, but the force that can be generated by the movable blade part **61** is increased, and in the region in which the load that is applied to the movable blade part **61** is low, the amount of rotation of the movable blade part **61** is relatively increased. Thereby, the time consumed until the cutting of the wire W ends can be suppressed from lengthening, and a time required for the whole binding operation can be shortened, as compared with the related art.
(221) Further, in the operation of cutting the wire W whose cross-sectional shape is circular, the load becomes highest immediately before the wire that the blade part has reached a position of a diameter is cut. Therefore, in the configuration where the two wires W aligned in parallel are cut, a phase difference is provided for timings at which the cuttings of the wires W are started. First, after starting the cutting of the first wire W, when the wire W is cut to a position of a half or more in the radial direction, the cutting of the second wire W is started.
(222) As compared with a case where two wires W aligned in parallel are cut at the same time, cutting one wire W reduces the load. Thereby, the load is reduced by starting the cutting of one wire W in advance. In addition, after the first wire W is cut to the position of a half or more in the radial direction and therefore the position where the load is the highest is passed, the cutting of the second wire W is started. Thereby, even when the two wires W are cut, the load is reduced. Further, the cutting of the second wire W is started before the cutting of the first wire W is completed. Thereby, an increase in time required for the cutting is suppressed.
(223) Further, when the sleeve **71** is moved in the forward direction denoted with the arrow A**1** by the operation of cutting the wire W wound on the reinforcing bars S, and as shown in FIG. **3**C, the opening/closing pin **71***a* is moved to the range in which it is located at the unlocking portion **73***c* of the opening/closing guide hole **73**L, the second side hook **70**L becomes movable in the direction away from the center hook **70**C by a predetermined amount.
(224) As described above, in the operation of feeding the wire W in the reverse direction and

winding the wire on the reinforcing bars S, the tip end side of the wire W needs to be locked in such a manner that the wire does not come off between the second side hook **70**L and the center hook **70**C. On the other hand, a reactive force of the force for pressing the wire W against the center hook **70**C with the second side hook **70**L is applied to the sleeve **71**, and this reactive force becomes the load that is applied to the rotary shaft **72** that moves and rotates the sleeve **71** and to the motor **80** connected to the rotary shaft **72** via the decelerator **81**.
(225) Therefore, the second side hook **70**L is provided with the locking portion **73***b* and the unlocking portion **73***c* in the opening/closing guide hole **73**L, and in the operation of winding the wire W on the reinforcing bars S, the sleeve **71** is moved to the position where the opening/closing pin **71***a* faces the locking portion **73***b* of the opening/closing guide hole **73**L, and after the wire W is wound on the reinforcing bars S, the sleeve **71** is moved to the position where the opening/closing pin **71***a* faces the unlocking portion **73***c* of the opening/closing guide hole **73**L.
(226) Thereby, in the operation of winding the wire W on the reinforcing bars S, the tip end side of the wire W can be locked in such a manner that the wire does not come off between the second side hook **70**L and center hook **70**C. In addition, after winding the wire W on the reinforcing bars S, the second side hook **70**L becomes movable in the direction away from the center hook **70**C by a predetermined amount, the reactive force of the force of pressing the wire W against the center hook **70**C with the second side hook **70**L is reduced, and the load that is applied to the motor **80** is reduced.
(227) By driving the motor **80** in the forward rotation direction, the sleeve **71** is moved in the forward direction denoted with the arrow A**1**, so that the bent portions **71***c***1** and **71***c***2** are moved toward the reinforcing bars S almost simultaneously with the cutting of the wire W as described above. Thereby, the tip end side of the wire W locked by the center hook **70**C and the second side hook **70**L is pressed toward the reinforcing bars S and bent toward the reinforcing bars S at the locking position as a fulcrum by the bending portion **71***c***1**. The sleeve **71** is further moved in the forward direction, so that the wire W locked between the second side hook **70**L and the center hook **70**C is maintained sandwiched by the bending portion **71***c***1**.
(228) In addition, the terminal end side of the wire W locked by the center hook **70**C and the first side hook **70**R and cut by the cutting unit **6** is pressed toward the reinforcing bars S and bent toward the reinforcing bars S at the locking position as a fulcrum by the bending portion **71***c***2**. The sleeve **71** is further moved in the forward direction, so that the wire W locked between the first side hook **70**R and the center hook **70**C is maintained sandwiched by the bending portion **71***c***2**.
(229) After bending the tip end side of the wire W and the terminal end side after the cutting toward the reinforcing bars S, the motor **80** is further driven in the forward rotation direction, so that the sleeve **71** is further moved in the forward direction. When the sleeve **71** is moved to a predetermined position and therefore reaches the operation region in which the wire W locked by the locking member **70** is twisted, the locking of the rotation regulation blade **74***a* is released.
(230) Thereby, the motor **80** is further driven in the forward rotation direction, so that the sleeve **71** is rotated in conjunction with the rotary shaft **72** and the wire W locked by the locking member **70** is twisted.
(231) In the second operation region in which the sleeve **71** is rotated to twist the wire W, the binding unit **7** twists the wire W locked by the locking member **70**, so that a force of pulling the sleeve **71** forward along the axis direction of the rotary shaft **72** is applied. On the other hand, when a force to move the sleeve **71** forward along the axis direction is applied, the rotary shaft **72** moves forward while receiving a force pushed backward by the spring **72***c*, and twists the wire W while moving forward.
(232) Therefore, the wire W is twisted while the locking member **70**, the sleeve **71**, and the rotary shaft **72** are moved forward with receiving the force pushed backward by the spring **72***c*, and therefore, a gap between the twisted portion of the wire W and the reinforcing bar S becomes small and the wire is brought into close contact with the reinforcing bar S along the reinforcing bar S.

Thereby, the slack before twisting the wire W can be removed, and the reinforcing bars S can be bound in a state where the wire W is in close contact with the reinforcing bars S.
(233) When it is detected that the load that is applied to the motor **80** is maximized as the wire W is twisted, the forward rotation of the motor **80** is stopped. Next, when the motor **80** is driven in the reverse rotation direction, the rotary shaft **72** is reversely rotated and the sleeve **71** is reversely rotated in conjunction with the reverse rotation of the rotary shaft **72**, the rotation regulation blade **74***a* is locked, so that the rotation of the sleeve **71** in conjunction with the rotation of the rotary shaft **72** is regulated. Thereby, the sleeve **71** is moved in the direction of the arrow A**2**, which is a backward direction.
(234) When the sleeve **71** is moved in the backward direction, the bending portions **71***c***1** and **71***c***2** are away from the wire W, and the holding of the wire W by the bending portions **71***c***1** and **71***c***2** is released. In addition, when the sleeve **71** is moved in the backward direction, the opening/closing pin **71***a* passes through the opening/closing guide holes **73**R and **73**L. Thereby, the first side hook **70**R is moved away from the center hook **70**C by the rotating operation about the shaft **71***b* as a fulcrum. In addition, the second side hook **70**L is moved away from the center hook **70**C by the rotating operation about the shaft **71***b* as a fulcrum. Thereby, the wire W comes off from the locking member **70**.
(235) Note that, as in the opening/closing guide hole **73**L of the modified embodiments shown in FIGS. **3**D to **3**F, in the configuration where the opening/closing guide hole **73**L is provided with the second locking portion **73***d*, when the sleeve **71** is further moved in the forward direction to a position where the operation of twisting the wire W becomes possible, the opening/closing pin **71***a* is located at the second locking portion **73***d* of the opening/closing guide hole **73**L. Thereby, even when the force by which the wire W is twisted is applied to the wire W, the wire W is suppressed from coming off between the second side hook **70**L and the center hook **70**C.
(236) Modified Embodiment of Implementation of Transmission Unit
(237) FIGS. **8**A to **8**C are side views showing a modified embodiment of the transmission unit of the present embodiment, and FIGS. **9**A to **9**C are side cross-sectional views showing the modified embodiment of the transmission unit of the present embodiment. Next, a transmission unit **9**B of the modified embodiment of the present embodiment will be described with reference to each drawing.
(238) The transmission unit **9**B includes a cutter lever **95** configured to rotate by an operation of the binding unit **7**, and a link **91** configured to connect the cutter lever **95** and the movable blade part **61**. The transmission unit **9**B is configured to transmit an operation of the binding unit **7** to the cutter lever **95** and the movable blade part **61** of the cutting unit **6** via the link **91**.
(239) The transmission unit **9**B is supported so that the cutter lever **95** can rotate about the shaft **90***b* as a fulcrum. The shaft **90***b* is attached to the frame **10***a* attached to the inside of the main body part **10**.
(240) The cutter lever **95** is an example of the displacement member, and includes a first cutter lever **95***a* and a second cutter lever **95***b* connected to the sleeve **71** via the moving member **75**. The cutter lever **95** is configured such that the first cutter lever **95***a* is engaged with the first engaging portion **75***b* provided to the moving member **75** and the second cutter lever **95***b* is engaged with the second engaging portion **75***c* provided to the moving member **75**.
(241) The cutter lever **95** is configured such that a length from an action point, which is the second connection portion connected to the sleeve **71**, to be pushed by the moving member **75** configured to move in conjunction with the sleeve **71** to the shaft **90***b* is different in the first cutter lever **95***a* and the second cutter lever **95***b*. The length from the shaft **90***b* to the action point to be pushed by the moving member **75** is configured to be longer in the second cutter lever **95***b* than in the first cutter lever **95***a*.
(242) That is, the length from the second engaging portion **75***c*, which is the action point to be pushed by the moving member **75** in the second cutter lever **95***b*, to the shaft **90***b* is configured to

be greater than the length from the first engaging portion **75***b*, which is the action point to be pushed by the moving member **75** in the first cutter lever **95***a*, to the shaft **90***b*.

(243) When the moving member **75** is moved in the forward direction in conjunction with the sleeve **71** moving in the forward direction denoted with the arrow A**1**, first, the first engaging portion **75***b* is engaged with the first cutter lever **95***a*. When the sleeve **71** is further moved in the forward direction denoted with the arrow A**1**, the second engaging portion **75***c* is engaged with the second cutter lever **95***b*. Further, the engagement between the first cutter lever **95***a* and the first engaging portion **75***b* is released.

(244) As for the link **91**, an end portion in the forward direction denoted with the arrow A**1** is connected to the movable blade part **61**, and an end portion in the backward direction denoted with the arrow A**2** is connected to the cutter lever **95**.

(245) Next, operations of the transmission unit **9**B are described. When the sleeve **71** is moved in the forward direction denoted with the arrow A**1**, the moving member **75** is moved in the forward direction denoted with the arrow A**1** in conjunction with the sleeve **71**. As shown in FIG. **9**B, the first engaging portion **75***b* is engaged with the first cutter lever **95***a* by the moving operation of the moving member **75** in the forward direction denoted with the arrow A**1**.

(246) When the moving member **75** is further moved in the forward direction denoted with the arrow A**1**, the cutter lever **95** is rotated in the direction of the arrow C**1** about the shaft **90***b* as a fulcrum with a ratio corresponding to the length from the shaft **90***b* to the action point pushed by the first engaging portion **75***b* of the moving member **75** in the first cutter lever **95***a* with respect to the amount of movement of the sleeve **71**.

(247) When the cutter lever **95** is rotated in the direction of the arrow C**1**, the rotating operation of the cutter lever **95** is transmitted to the movable blade part **61** via the link **91**, so that the movable blade part **61** is rotated in the direction of the arrow D**1**. Therefore, the movable blade part **61** is rotated in the direction of the arrow D**1** by the moving operation of the sleeve **71** in the forward direction, so that cutting of the wire W is started.

(248) When the sleeve **71** is further moved in the forward direction denoted with the arrow A**1**, the second engaging portion **75***c* of the moving member **75** is engaged with the second cutter lever **95***b*, as shown in FIG. **8**C. Thereby, the cutter lever **95** is rotated in the direction of the arrow C**1** about the shaft **90***b* as a fulcrum with a ratio corresponding to the length from the shaft **90***b* to action point pushed by the second engaging portion **75***c* of the moving member **75** in the second cutter lever **95***b* with respect to the amount of movement of the sleeve **71**. Further, the engagement between the first cutter lever **95***a* and the first engaging portion **75***b* is released.

(249) The duration for which the first cutter lever **95***a* and the first engaging portion **75***b* are engaged is a duration from when the movable blade part **61** starts rotation in the cutting unit **6** until the cutting of the first wire W is started. In addition, the duration for which the second cutter lever **95***b* and the second engaging portion **75***c* are engaged is a duration from when the movable blade part **61** is further rotated in the cutting unit **6** and the cutting of the first wire W is started until the cutting of the second wire W ends.

(250) The cutter lever **95** is configured such that the length from the shaft **90***b* to the action point pushed by the moving member **75** is longer in the second cutter lever **95***b* than in the first cutter lever **95***a*. Thereby, while the first cutter lever **95***a* and the first engaging portion **75***b* are engaged, the amount of rotation of the movable blade part **61** becomes relatively large with respect to the amount of movement of the sleeve **71** that rotates the cutter lever **95**.

(251) On the other hand, since the cutting of the wire W is not started while the first cutter lever **95***a* and the first engaging portion **75***b* are engaged, the increase in load that is applied to the movable blade part **61** is suppressed, and the increase in load that is applied to the cutter lever **95** connected to the movable blade part **61** via the link **91** is suppressed.

(252) Since the cutter lever **95** is connected to the sleeve **71** via the moving member **75**, the increase in load that is applied to the cutter lever **95** is suppressed, so that the increase in load that

is applied to the rotary shaft **72** that moves the sleeve **71** and to the motor **80** connected to the rotary shaft **72** via the decelerator **81** is suppressed.
(253) Therefore, in the region in which the load is low until the cutting of the first wire W is started, a time consumed to rotate the movable blade part **61** to a position where the cutting of the wire W is started can be shortened by relatively increasing the amount of rotation of the movable blade part **61**.
(254) While the second cutter lever **95***b* and the second engaging portion **75***c* are engaged, the amount of rotation of the movable blade part **61** becomes relatively small with respect to the amount of movement of the sleeve **71** that rotates the cutter lever **95**. On the other hand, since the length from the shaft **90***b* to the action point pushed by the moving member **75** is configured to be longer in the second cutter lever **95***b* than in the first cutter lever **95***a*, the force that can be generated by the movable blade part **61** from the cutter lever **95** via the link **91** increases.
(255) When the cutting of the first wire W is started, the load that is applied to the movable blade part **61** increases. On the other hand, the force that can be generated by the movable blade part **61** increases, so that the load that is applied to the movable blade part **61** is canceled and the increase in load that is applied to the cutter lever **95** connected to the movable blade part **61** via the link **91** is suppressed.
(256) The increase in load that is applied to the cutter lever **95** is suppressed, so that the increase in load that is applied to the rotary shaft **72** that moves the sleeve **71** and to the motor **80** connected to the rotary shaft **72** via the decelerator **81** is suppressed.
(257) Therefore, in a region in which the load is high from when the cutting of the first wire W is started until the cutting of the second wire W ends, the increase in load that is applied to the motor **80** can be suppressed by increasing the force that can be generated by the movable blade part **61**. In addition, in the region in which the load is high, the amount of rotation of the movable blade part **61** becomes relatively small, but in the region in which the load is low, the time consumed until the cutting of the wire W ends can be suppressed from lengthening by relatively increasing the amount of rotation of the movable blade part **61**.
(258) Note that, in the above embodiment, the cutter lever **75** has such a configuration that whether the first engaging portion **75***b* of the moving member **75** and the first cutter lever **95***a* are engaged or whether the second engaging portion **75***c* of the moving member **75** and the second cutter lever **95***b* are engaged is switched by the rotating operation of the cutter lever **85** about the shaft **90***b* as a fulcrum, and therefore, the length from the shaft **90***b* to the first connection portion connected to the sleeve **71** is switched.
(259) Thereby, the cutter lever **95** makes it possible to switch the amount of rotation (amount of movement) of the movable blade part **61** and the force that can be generated by the movable blade part **61**, within the rotating range (moving range) of the movable blade part **61**.
(260) On the other hand, the cutter lever **95** may have such a configuration that the portion to which the link **91** is connected can be switched by the rotating operation of the cutter lever **85** about the shaft **90***b* as a fulcrum, and therefore, the length from the shaft **90***b* to the second connection portion connected to the link **91** can be switched.

Claims

1. A binding machine comprising: a wire feeding unit comprising a pair of gears configured to feed a wire; a curl forming unit comprising a curl guide configured to form a path along which the wire fed by the wire feeding unit is to be wound around an object; a cutting unit comprising blade parts configured to cut the wire wound on the object; a binding unit including a locking member configured to lock the wire, a sleeve configured to actuate the locking member, and a rotary shaft configured to move and rotate the sleeve, the binding unit being configured to twist the wire wound on the object and cut by the cutting unit; and a drive unit configured to drive the binding unit,

wherein the drive unit comprises a motor and a decelerator configured to perform deceleration and amplification of torque, wherein the decelerator comprises: a sun gear, a planetary gear in mesh with the sun gear, a planet cage configured to support the planetary gear, and an internal gear in mesh with the planetary gear, and wherein the planet cage includes a front side portion protruding from the internal gear along an axis direction and a rear side portion located inside the internal gear along the axis direction, wherein the front side portion and the rear side portion of the planet cage are rotatably supported, and wherein the rear side portion is rotatably supported on the internal gear.

2. The binding machine according to claim 1, wherein the planet cage is configured such that one side along the axis direction is rotatably supported by a main body part to which the internal gear is fixed and the other side is rotatably supported by the internal gear.

3. The binding machine according to claim 1, wherein the planet cage is configured such that one side along the axis direction is supported on an inner side of the internal gear.

4. A binding machine comprising: a wire feeding unit comprising a pair of gears configured to feed a wire; a curl forming unit comprising a curl guide configured to form a path along which the wire fed by the wire feeding unit is to be wound around an object; a cutting unit comprising blade parts configured to cut the wire wound on the object; a binding unit including a locking member configured to lock the wire, a sleeve configured to actuate the locking member, and a rotary shaft configured to move and rotate the sleeve, the binding unit being configured to twist the wire wound on the object and cut by the cutting unit; and a drive unit configured to drive the binding unit, wherein the drive unit comprises a motor and a decelerator configured to perform deceleration and amplification of torque, wherein the decelerator comprises: a first sun gear attached to a shaft of the motor, a first planetary gear in mesh with the first sun gear, a first planet cage configured to support the first planetary gear, a second sun gear provided to the first planet cage, a second planetary gear in mesh with the second sun gear, an internal gear in mesh with the second planetary gear, and a second planet cage configured to support the second planetary gear, wherein the second planet cage is-includes a front side portion protruding from the internal gear along an axis direction and a rear side portion located inside the internal gear along the axis direction, wherein the front side portion and the rear side portion of the second planet cage are rotatably supported, and wherein the rear side portion is rotatably supported on the internal gear.

5. The binding machine according to claim 4, wherein the decelerator comprises a gear holder between the first planet cage and the second planetary gear.